

Maharaja Surajmal Institute of Technology

MSIT – Comprehensive Institutional Reference Document

West Delhi, New Delhi – 110058 | Affiliated: GGSIPU | Approved: AICTE

1. About Maharaja Surajmal Institute of Technology (MSIT)

Maharaja Surajmal Institute of Technology (MSIT) is a premier engineering college located in Janakpuri, West Delhi, India. Established in 2001, the institute is affiliated with Guru Gobind Singh Indraprastha University (GGSIPU) and is approved by the All India Council for Technical Education (AICTE).

MSIT is named after the legendary Jat king Maharaja Surajmal, who was known for his valor, administrative brilliance, and patronage of education. The institution upholds this legacy by nurturing technically skilled and ethically grounded engineers.

MSIT is recognized for its strong academic foundation, experienced and research-active faculty, industry-aligned curriculum, and robust emphasis on both technical competence and professional development. The institute has consistently produced graduates who have gone on to top companies and postgraduate programs globally.

The campus spans a significant area in Janakpuri, housing state-of-the-art labs, a well-stocked central library, digital classrooms, and student activity zones that encourage holistic growth.

1.1 Vision and Mission

Vision: To be a center of excellence in technical education, nurturing innovative engineers who contribute meaningfully to society and the global technology landscape.

Mission:

- Provide high-quality technical education aligned with industry needs and global standards.
- Foster a culture of research, innovation, and entrepreneurship among students and faculty.
- Promote ethical values, teamwork, and leadership through co-curricular and extracurricular activities.
- Build strong industry-academia partnerships to enhance placement outcomes and real-world readiness.

1.2 Accreditation and Recognition

- Affiliated with Guru Gobind Singh Indraprastha University (GGSIPU), New Delhi.
- Approved by the All India Council for Technical Education (AICTE).
- Recognized by the Government of Delhi.
- ISO-certified quality management system for academic processes.
- Active member of Computer Society of India (CSI) and IEEE student chapters.

2. Courses Offered

MSIT offers Bachelor of Technology (B.Tech) four-year undergraduate programs across four engineering disciplines. Each program is designed to provide in-depth theoretical knowledge and practical skills through laboratory work, projects, and industry internships.

Program	Branch Code	Intake (Seats)	Duration
B.Tech – Computer Science & Engineering	CSE	120	4 Years (8 Semesters)
B.Tech – Information Technology	IT	60	4 Years (8 Semesters)
B.Tech – Electronics & Communication Engg.	ECE	60	4 Years (8 Semesters)
B.Tech – Electrical Engineering	EE	60	4 Years (8 Semesters)

2.1 Program Highlights

- CSE: Focus on software engineering, AI/ML, cloud computing, cybersecurity, and full-stack development.
- IT: Emphasis on networking, database systems, web technologies, and enterprise solutions.
- ECE: Covers VLSI design, embedded systems, signal processing, and communication networks.
- EE: Includes power systems, control engineering, electric drives, and renewable energy.

3. Admission Process

Admissions to MSIT B.Tech programs are conducted through the centralized counseling process managed by Guru Gobind Singh Indraprastha University (GGSIPU). The entire process is transparent, merit-based, and follows guidelines set by the university and AICTE.

3.1 Eligibility Criteria

- Candidates must have passed Class 12 (10+2) or equivalent from a recognized board.
- Minimum 60% aggregate marks in Physics, Chemistry, and Mathematics (PCM).
- Must have qualified JEE Main (conducted by NTA) in the respective admission year.
- Age limit: Minimum 17 years as on 31st December of the admission year.

3.2 Selection Process

- Step 1: Appear and qualify JEE Main exam conducted by National Testing Agency (NTA).
- Step 2: Register on the GGSIPU admissions portal (ipu.admissions.nic.in).

- Step 3: Fill preferences for institute and branch during online counseling.
- Step 4: Seat allotment based on JEE Main rank and category (General / OBC / SC / ST / PH).
- Step 5: Report to the allotted institute with original documents for verification and fee payment.

3.3 Reservation Policy

Category	Reservation Percentage
General (Open Merit)	27%
OBC (Other Backward Class)	27%
SC (Scheduled Caste)	15%
ST (Scheduled Tribe)	7.5%
EWS (Economically Weaker Section)	10%
PH / Differently Abled (Horizontal)	3% (across all categories)

4. Fee Structure (Approximate – Academic Year 2024-25)

Fee Component	Amount (Per Year)	Remarks
Tuition Fees	₹1,20,000 – ₹1,50,000	Payable per semester
Development Fund	₹10,000 – ₹15,000	Annual
Library & Laboratory Charges	₹8,000 – ₹12,000	Annual
Exam & Registration Fees	₹5,000 – ₹8,000	University fee
Misc. Charges	₹5,000 – ₹10,000	Sports, culture, etc.
Total (Approx.)	₹1,48,000 – ₹1,95,000	Subject to revision

Note: Scholarship and fee waivers are available for SC/ST students and meritorious candidates as per GGSIPU and Delhi Government norms. Students may also apply for the Central Sector Scholarship and other national-level schemes.

5. B.Tech Curriculum Overview

The B.Tech curriculum at MSIT is designed as per GGSIPU guidelines and is regularly updated to align with industry trends. The program is structured into 8 semesters over 4 years, progressively moving from foundational to advanced topics, culminating in a major project and industrial training.

5.1 Year 1 – Foundation Courses

Semester	Key Subjects	Credits
Sem 1	Engineering Mathematics-I, Engineering Physics, Basic Electrical Engineering, Introduction to Computing, Environmental Science	20
Sem 2	Engineering Mathematics-II, Engineering Chemistry, Programming in C, Engineering Graphics, Communication Skills	20

5.2 Year 2 – Core Engineering

Semester	Key Subjects	Credits
Sem 3	Data Structures, Digital Electronics, Discrete Mathematics, OOP with Java, Probability & Statistics	22
Sem 4	Design & Analysis of Algorithms, Computer Organization, Signals & Systems, Software Engineering, Database Management Systems	22

5.3 Year 3 – Specialization

Semester	Key Subjects	Credits
Sem 5	Operating Systems, Computer Networks, Theory of Computation, Elective-I (AI/ML/IoT), Minor Project-I	22
Sem 6	Compiler Design, Distributed Systems, Elective-II, Open Elective-I, Minor Project-II	22

5.4 Year 4 – Industry Integration

Semester	Key Subjects	Credits
Sem 7	Elective-III, Elective-IV, Open Elective-II, Major Project Phase-I, Industrial Training (6 weeks)	22

Sem 8	Major Project Phase-II, Seminar, Comprehensive Viva	18
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6. Faculty and Heads of Departments

MSIT has a diverse and highly qualified faculty body. The institute promotes continuous learning, research publications, and faculty development programs. Below are the Heads of Department (dummy data for reference purposes).

Department	Head of Department	Qualification	Experience
Computer Science & Engineering (CSE)	Dr. Amit Sharma	Ph.D. (IIT Delhi)	22 Years
Information Technology (IT)	Dr. Neha Gupta	Ph.D. (NIT Trichy)	18 Years
Electronics & Communication Engg. (ECE)	Dr. Pradeep Sangwan	Ph.D. (Delhi University)	20 Years
Electrical Engineering (EE)	Dr. Sunita Mehta	Ph.D. (IIT Roorkee)	17 Years
Applied Sciences & Humanities	Dr. Pradeep Kumar	Ph.D. (JNU)	15 Years

Note: Faculty data above is dummy/representative. Actual faculty details can be found on the official MSIT website.

7. Student Academic Records (Dummy Data)

The following semester-wise SGPA records represent sample student profiles from the current batch. Data is illustrative and used for academic planning purposes only.

7.1 Student: Rahul Verma – B.Tech CSE (Enrollment: 2021-25)

Semester	SGPA	CGPA (Cumulative)	Rank in Class
Semester 1	8.2	8.20	14
Semester 2	8.5	8.35	10
Semester 3	8.7	8.46	8
Semester 4	8.9	8.57	5
Semester 5	9.0 (Projected)	8.66 (Projected)	—

7.2 Student: Priya Singh – B.Tech IT (Enrollment: 2021-25)

Semester	SGPA	CGPA (Cumulative)	Rank in Class
Semester 1	7.8	7.80	22
Semester 2	8.1	7.95	18
Semester 3	8.4	8.10	12
Semester 4	8.6	8.22	9

7.3 Student: Aman Gupta – B.Tech ECE (Enrollment: 2021-25)

Semester	SGPA	CGPA (Cumulative)	Rank in Class
Semester 1	7.5	7.50	30
Semester 2	7.9	7.70	26
Semester 3	8.2	7.86	20
Semester 4	8.3	7.97	18

8. Placement Cell and Career Development

The Training and Placement Cell (TPC) at MSIT plays a pivotal role in bridging the gap between students and the industry. The cell organizes pre-placement training, mock interviews, aptitude workshops, and soft skills development sessions throughout the year.

8.1 Placement Statistics (2023-24)

Metric	CSE / IT	ECE / EE
Average Package	₹7.5 LPA	₹5.5 LPA
Highest Package	₹22 LPA	₹14 LPA
Placement Percentage	~92%	~80%
Students Placed (Total)	150+	70+

8.2 Top Recruiters

- IT Giants: TCS, Infosys, Wipro, HCL Technologies, Tech Mahindra
- Product Companies: Amazon, Microsoft, Adobe, Paytm, Zomato, Flipkart
- Startups & Mid-Cap: Nagarro, GlobalLogic, Persistent Systems, Mphasis
- Core Engineering: L&T, Siemens, BHEL, ABB, Schneider Electric
- Consulting & Finance: Deloitte, EY, KPMG, Accenture

8.3 Pre-Placement Training

- Aptitude & Reasoning: Quantitative, Logical, and Verbal training starting from 3rd year.
- Technical Rounds: Data structures, system design, OOP concepts, DBMS, OS.
- Mock Interviews: Conducted with alumni and industry professionals.
- Resume Building: Personalized sessions for crafting ATS-friendly resumes.
- Communication Skills: Group discussions, public speaking, presentation workshops.

9. Campus Infrastructure

MSIT's campus in Janakpuri is designed to support an enriching academic and social experience for its students. The campus infrastructure is regularly upgraded to meet modern educational standards.

9.1 Academic Facilities

- Smart Classrooms: 40+ classrooms equipped with projectors, interactive boards, and high-speed internet.
- Computer Labs: 8 fully equipped labs with 500+ workstations running Windows and Linux environments.
- Electronics & Communication Lab: PCB design, oscilloscopes, signal generators, DSP kits.
- VLSI & Embedded Systems Lab: FPGA boards, microcontroller kits, IoT development platforms.
- Electrical Engineering Lab: Power electronics kits, motor drives, simulation software (MATLAB/SIMULINK).

9.2 Library

- Central Library with 40,000+ volumes covering all engineering disciplines.
- Access to IEEE Xplore, Springer, Elsevier, and GGSIPU digital library resources.
- Dedicated reading halls with 200+ seating capacity.
- Digital catalog and OPAC system for easy book search and borrowing.

9.3 Other Facilities

- Wi-Fi Campus: High-speed internet (1 Gbps fiber) across all buildings.
- Cafeteria: Hygienic and affordable food court with multiple vendors.
- Sports Complex: Basketball, Volleyball, Badminton courts; Table Tennis room.
- Medical Room: On-campus first aid facility with visiting medical staff.
- Hostel: Limited hostel facility available for outstation students (subject to availability).
- Transport: College bus services covering major routes across Delhi-NCR.

10. Student Activities and Clubs

MSIT encourages students to participate in technical and cultural activities beyond academics. The institute hosts several student-run clubs and annual events.

10.1 Technical Clubs

- CodeForge Club: Competitive programming, hackathons, and coding contests.
- IEEE Student Branch: Technical workshops, seminars, and paper presentations.
- Robotics & IoT Club: Hardware projects, drone competitions, and maker events.
- Cyber Security Club: CTF competitions, ethical hacking workshops.
- Data Science & AI Club: ML projects, Kaggle competitions, AI seminars.

10.2 Cultural & Sports Events

- Udaan: Annual cultural fest with music, dance, drama, and fine arts.
- TechFest: Annual technical symposium featuring paper presentations and project expos.
- Sports Meet: Inter-college and intra-college tournaments in cricket, football, and athletics.
- NSS Chapter: National Service Scheme activities including blood donation drives and community outreach.

11. Contact Information

Field	Details
Institute Name	Maharaja Surajmal Institute of Technology (MSIT)
Address	C-4, Janakpuri, New Delhi – 110058
Website	www.msit.in
Email	info@msit.in
Phone	+91-11-25507788
University	Guru Gobind Singh Indraprastha University (GGSIPU)
Approval Body	All India Council for Technical Education (AICTE)

Note: This document contains dummy/representative data for project and RAG pipeline testing purposes. Official information should be verified at www.msit.in or the GGSIPU website.